Poisoning Deaths in South Africa

2026-03-09

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Overview

Poisoning is a significant cause of injury death in South Africa. This page describes the epidemiology of poisoning deaths recorded in the national vital registration (VR) system, using underlying-cause ICD-10 codes. Four intent categories are distinguished, though **the category boundaries are substantially distorted by the absence of a manner of death field on the South African death notification form** (see methodological note below):

ICD-10 block	Category	Description
X40–X49	Accidental / unspecified	Unintentional exposure or intent not documented on DNF (ICD-10 default)
X60–X69	Self-harm	Intentional self-poisoning
X85	Assault	Poisoning inflicted by another person
Y10–Y19	Undetermined	Intent unknown or not specified

! Caution: intent categories reflect ICD-10 coding defaults, not certified intent

South Africa's death notification form (DNF) **does not include a field for manner of death**, contrary to the WHO-recommended Medical Certificate of Cause of Death (1). Under ICD-10 Volume 2 coding rules, when manner of death is not stated by the certifier, external cause deaths are assigned to the most appropriate **Accidental** code by default (2).
This structural omission has two direct consequences for the intent breakdown presented in this analysis:

- **"Accidental" (X40–X49) is substantially over-represented.** In the absence of a certified manner, Stats SA coders apply the ICD-10 default and assign an accidental poisoning code. The scale of this effect is illustrated starkly for firearm deaths: Groenewald et al. (2023) showed that in Stats SA's 2017 data, **98.7% of firearm deaths were coded as accidental**, while the National Cause-of-Death Validation (NCoDV) forensic data from the same period found 88.5% were homicides (3). The same mechanism applies to poisoning, where the absence of certified intent causes genuine self-harm and assault deaths to be silently absorbed into the accidental category.
- **"Self-harm" (X60–X69) and "Undetermined" (Y10–Y19) are substantially under-represented.** These categories require the certifier to explicitly document suicidal intent or uncertainty — information that in South Africa is almost exclusively available from forensic pathology reports. Non-forensic certifiers (the majority for poisoning deaths in clinical settings) default to the accidental category or do not specify intent, regardless of clinical context or stigma considerations (4–6).

Interpretation: The "Accidental" category in these data should be read as **"accidental or intent not specified."** The true proportion of intentional and undetermined-intent poisoning deaths is almost certainly higher than shown. Accurate intent-stratified estimates require triangulation with forensic mortality surveillance data, poison information centre (NPIS) call records, and other independent sources (7).

Summary statistics

Table 1: Table 1. Overall summary — poisoning deaths, vital registration

Metric	Value
Total poisoning deaths	62,399
Analysis years	1997 - 2022
% Male	59.3%
Median age (years)	30
% Accidental/unspec.	43.3%
% Self-harm	4.2%
% Assault	0%

Metric	Value
% Undetermined	52.5%

Trends over time

Annual deaths and crude death rate

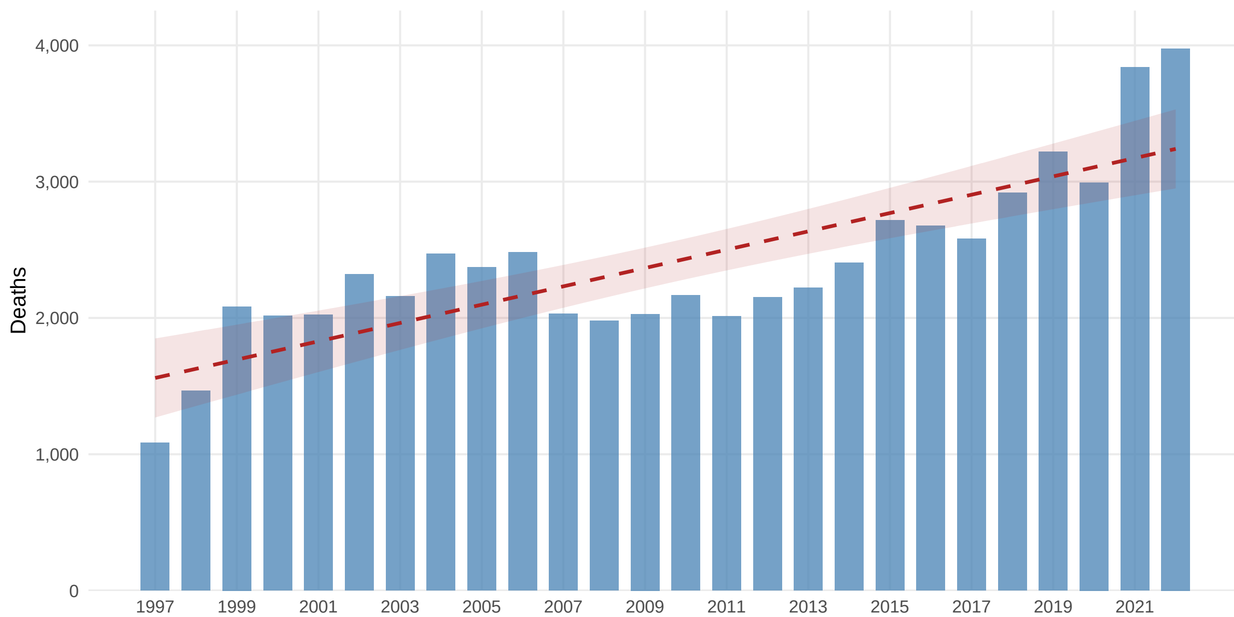

Figure 1: Fig. 1. Annual poisoning deaths. Bars = observed count; dashed red line = OLS trend with 95% CI.

Deaths by intent

The **"Accidental/unspec."** category (blue) dominates because ICD-10 Vol. 2 coding rules assign an accidental code whenever manner of death is not stated — see the methodological note above (1,2).

Annual summary

Table 2: Table 2. Annual poisoning deaths by intent, % male, and median age.

Year	Total	Accidental/unspec.	Self-harm	Assault	Undetermined	% Male	Median age
1,997	1,085	601	21	0	463	58.7	26

Year	Total	Accidental/unspec.	Self-harm	Assault	Undetermined	% Male	Median age
1,998	1,467	812	26	0	629	58.1	29
1,999	2,084	1,016	46	0	1,022	60.3	29
2,000	2,015	888	45	1	1,081	58.7	31
2,001	2,025	799	67	0	1,159	60.3	30
2,002	2,320	1,128	56	2	1,134	57.6	34
2,003	2,159	1,011	55	0	1,093	58.9	29
2,004	2,471	1,121	84	0	1,266	58.2	28
2,005	2,373	1,124	110	0	1,139	58.9	27
2,006	2,483	1,236	84	0	1,163	57.8	25
2,007	2,030	780	109	0	1,141	58.5	28
2,008	1,981	816	99	0	1,066	58.0	28
2,009	2,029	835	72	0	1,122	62.0	28
2,010	2,167	895	108	0	1,164	59.8	27
2,011	2,012	883	102	0	1,027	61.8	30
2,012	2,151	913	131	0	1,107	59.8	28
2,013	2,222	825	162	0	1,235	59.4	29
2,014	2,404	888	139	0	1,377	60.1	28
2,015	2,716	959	106	0	1,651	61.9	29
2,016	2,676	936	82	0	1,658	59.7	29
2,017	2,582	771	98	0	1,713	59.1	30
2,018	2,919	851	82	0	1,986	59.1	30
2,019	3,218	1,155	77	0	1,986	57.6	31
2,020	2,994	924	69	2	1,999	57.8	32
2,021	3,839	2,313	353	0	1,173	59.1	31
2,022	3,977	2,554	219	1	1,203	61.0	32

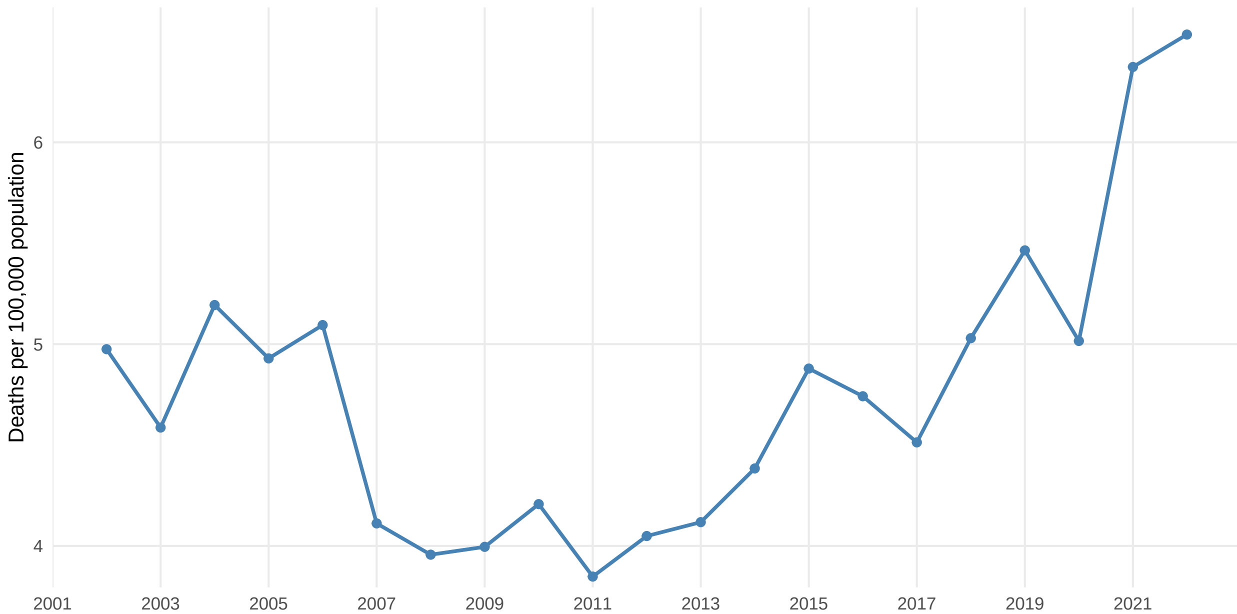

Figure 2: Fig. 2. Annual poisoning crude death rate (deaths per 100,000 population).

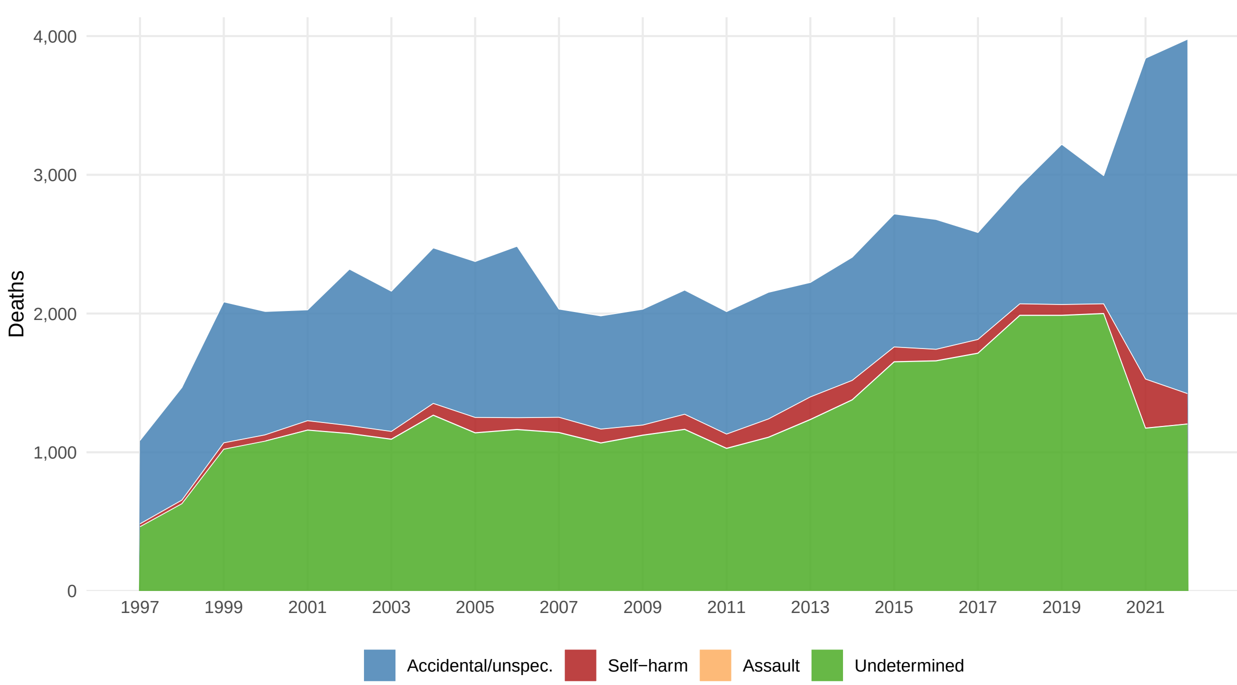

Figure 3: Fig. 3. Annual poisoning deaths by intent (stacked area). Blue = accidental/unspecified; red = self-harm; orange = assault; green = undetermined.

Epicurve

Monthly time series

The monthly epicurve shows all poisoning deaths from 1997 to 2022, stacked by intent category. The overlaid LOESS smoother (dark line with 95% CI) tracks the long-term trajectory of total counts while the stacked colours reveal whether any compositional shift in intent type accompanied the volume change.

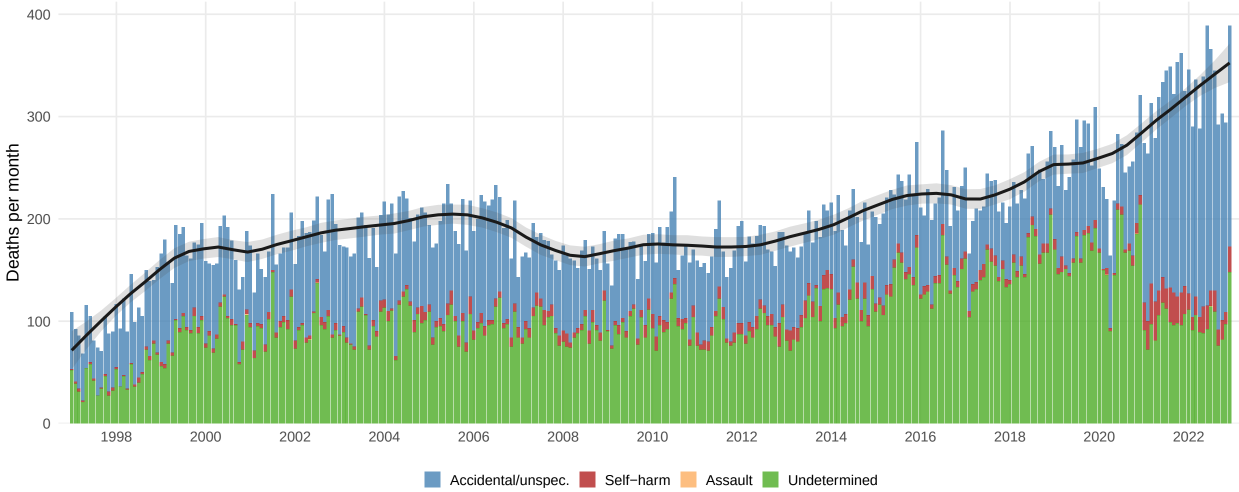

Figure 4: Fig. 3. Monthly poisoning deaths by intent (stacked bars). Dark line + band = LOESS total-count smoother (span 5-year window) with 95% CI. The dominant blue category ('Accidental/unspec.') is inflated by the ICD-10 coding default applied whenever manner of death is absent from the death notification form.

Monthly seasonality

Average deaths per calendar month (averaged across all years), with ±1 SD error bars. A consistent seasonal pattern would suggest environmental or behavioural drivers (e.g. agricultural pesticide exposure in summer, indoor heating–related gas poisoning in winter).

Age and sex

Age-sex pyramid

The population pyramid shows the sex- and age-distribution of all poisoning deaths. Males to the left, females to the right.

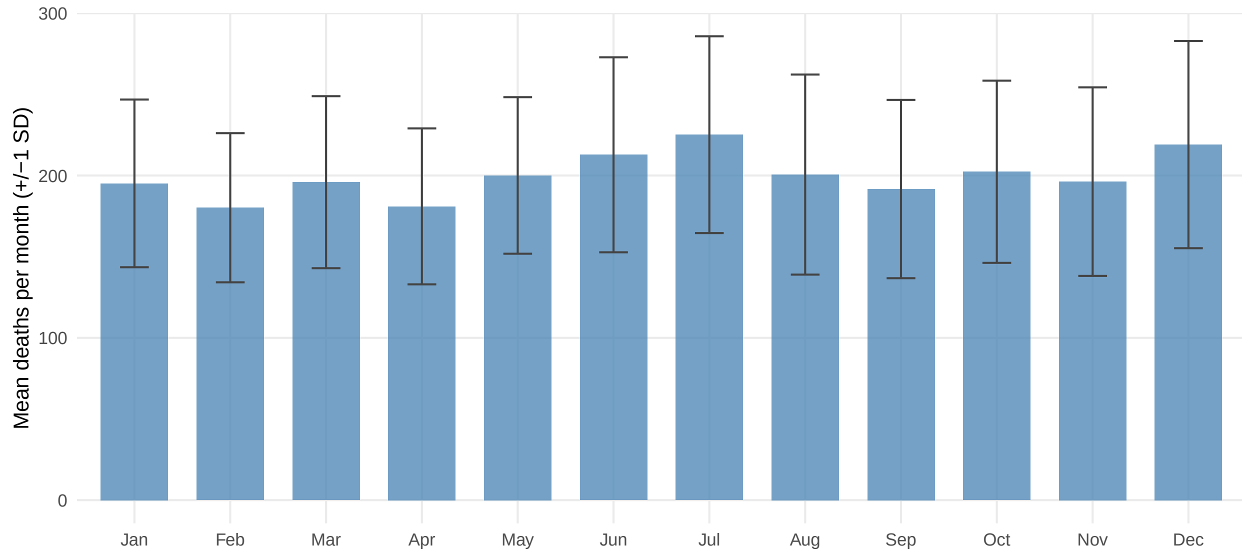

Figure 5: Fig. 4. Mean poisoning deaths per calendar month (+/-1 SD across years).

Age-sex pyramid by intent

Age × sex table

Table 3: Table 3. Poisoning deaths by age group and sex

Age group	Male	Female	Total	% Male
<1	4,502	3,874	8,376	53.7
1-4	2,394	2,023	4,417	54.2
5-9	693	565	1,258	55.1
10-14	558	832	1,390	40.1
15-19	1,424	2,825	4,249	33.5
20-24	2,969	2,406	5,375	55.2
25-29	3,904	2,143	6,047	64.6
30-34	4,018	1,820	5,838	68.8
35-39	3,518	1,598	5,116	68.8
40-44	3,009	1,304	4,313	69.8
45-49	2,558	1,252	3,810	67.1
50-54	2,046	1,014	3,060	66.9
55-59	1,581	835	2,416	65.4
60-64	1,285	684	1,969	65.3

Age group	Male	Female	Total	% Male
65-69	944	610	1,554	60.7
70-74	611	521	1,132	54.0
75+	1,005	1,074	2,079	48.3

Age-group trends over time

Age × year heatmap — nominal counts

Each cell shows the **absolute number of poisoning deaths** in that age band and calendar year. This reveals which groups contribute the most deaths in volume terms and allows direct comparison across time.

Age × year heatmap — proportional share

Each cell shows that age band's **share of that year's total poisoning deaths** (%). Using proportions rather than raw counts removes the effect of overall volume changes and isolates structural shifts in the age distribution — e.g. a darkening row signals a rising share for that group regardless of whether total deaths rose or fell.

Estimated annual percentage change (EAPC) by age group

The EAPC is derived from a log-linear regression of annual death counts on calendar year for each age band independently (with a 0.5 continuity correction for zero counts). It estimates the average year-on-year percentage change across 1997–2022. Red = increasing trend; blue = decreasing; faded points indicate $\underline{p}$ 0.05. This is the standard WHO / GBD approach for characterising mortality trends by subgroup.

Age-group trends by sex

The three figures below stratify the age-trend analysis by sex, enabling direct comparison between male and female poisoning death patterns — e.g. whether rising counts in older age groups are driven by males, and whether any increase in younger age groups is female-predominant.

Age × year heatmap — counts by sex

Two side-by-side heatmaps showing raw death counts. Allows comparison of absolute burden between sexes across the age-year space.

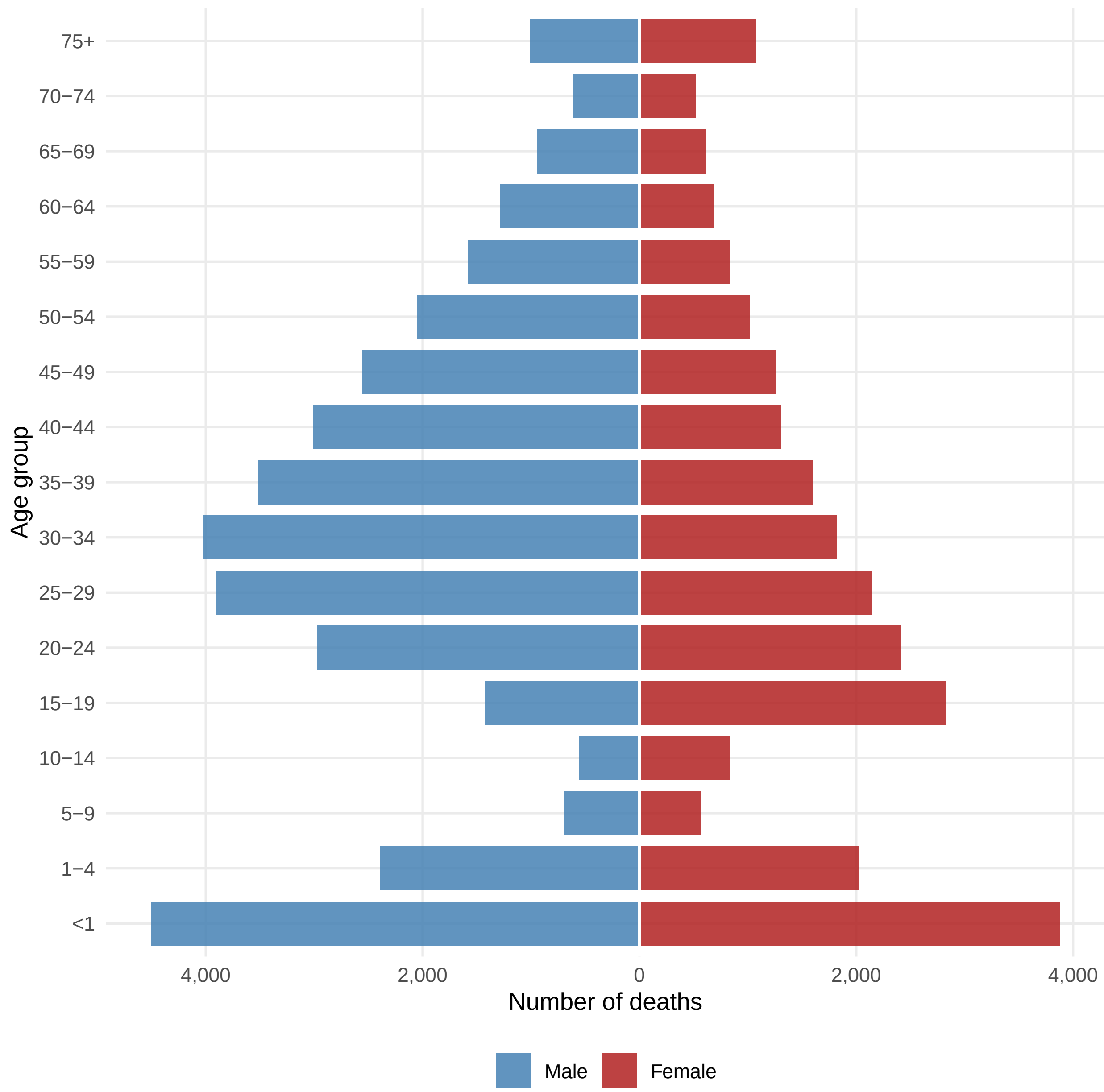

Figure 6: Fig. 5. Age-sex pyramid — all poisoning deaths. Blue = male; red = female.

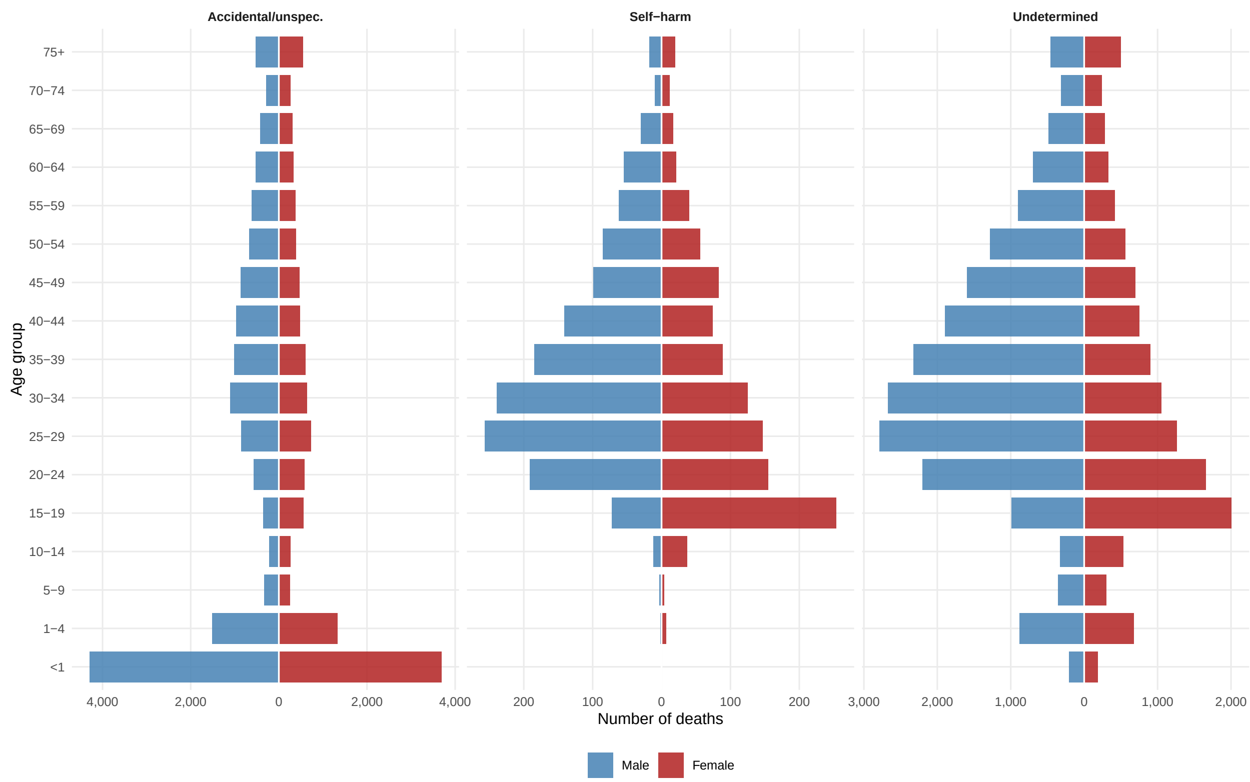

Figure 7: Fig. 6. Age-sex pyramids by intent category (accidental/unspecified, self-harm, undetermined). The 'Accidental/unspec.' panel is the most affected by the ICD-10 coding default and likely absorbs cases of genuine self-harm and undetermined intent. X-axis scales differ by panel.

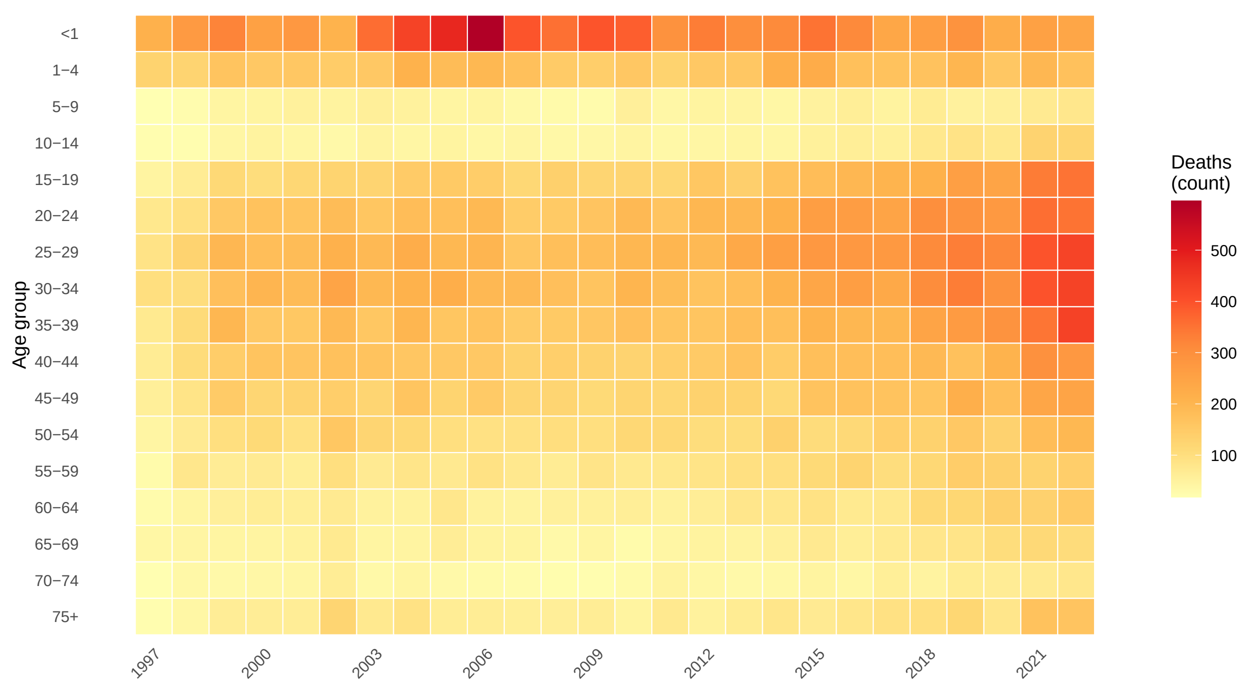

Figure 8: Fig. 9a. Annual poisoning deaths by age group (raw count). Darker = more deaths.

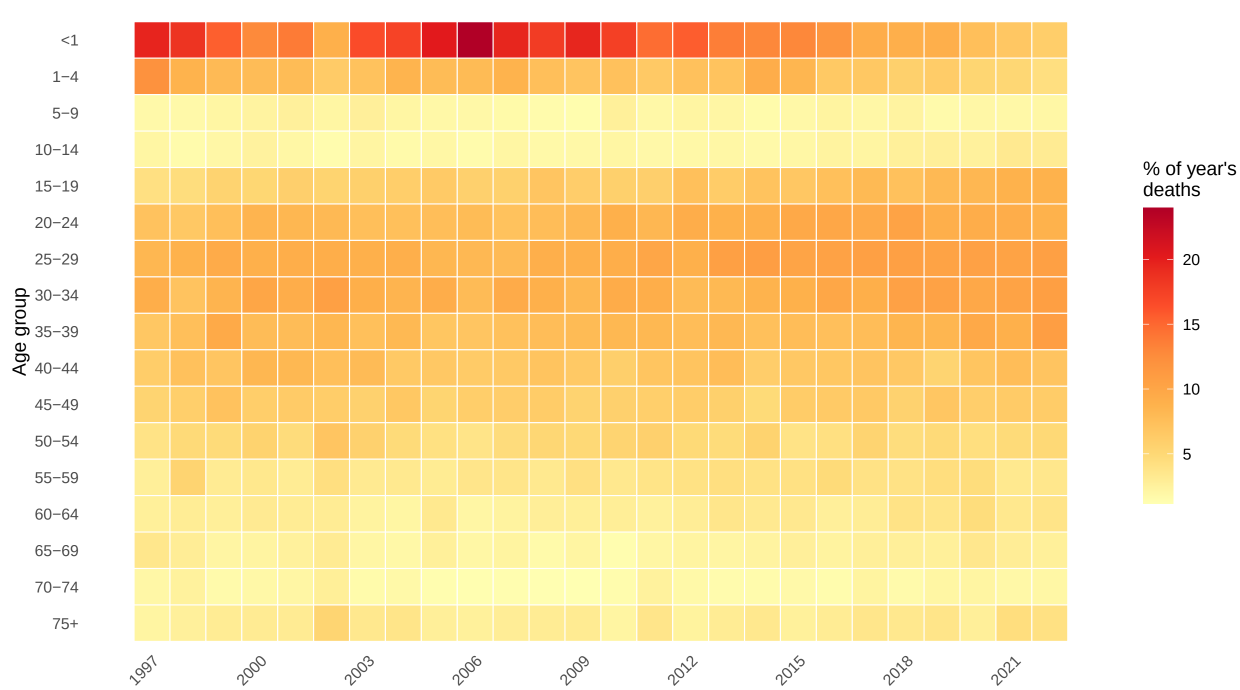

Figure 9: Fig. 9b. Proportion of annual poisoning deaths by age group (heatmap). Darker = higher share. Each column sums to 100%.

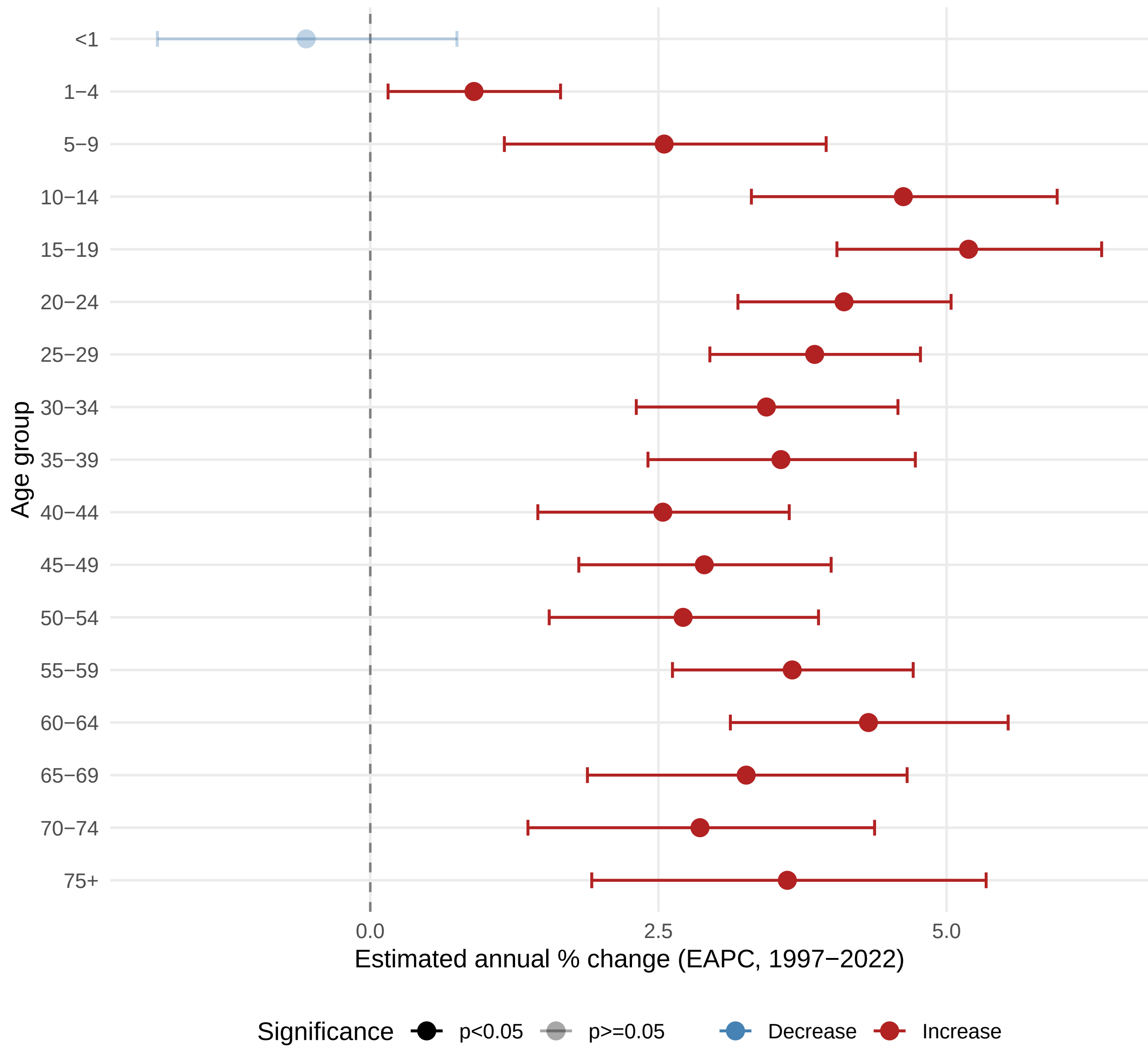

Figure 10: Fig. 10. EAPC by age group with 95% CI. Solid (opaque) points are statistically significant ($p < 0.05$). EAPC = estimated annual % change in poisoning death count, 1997–2022.

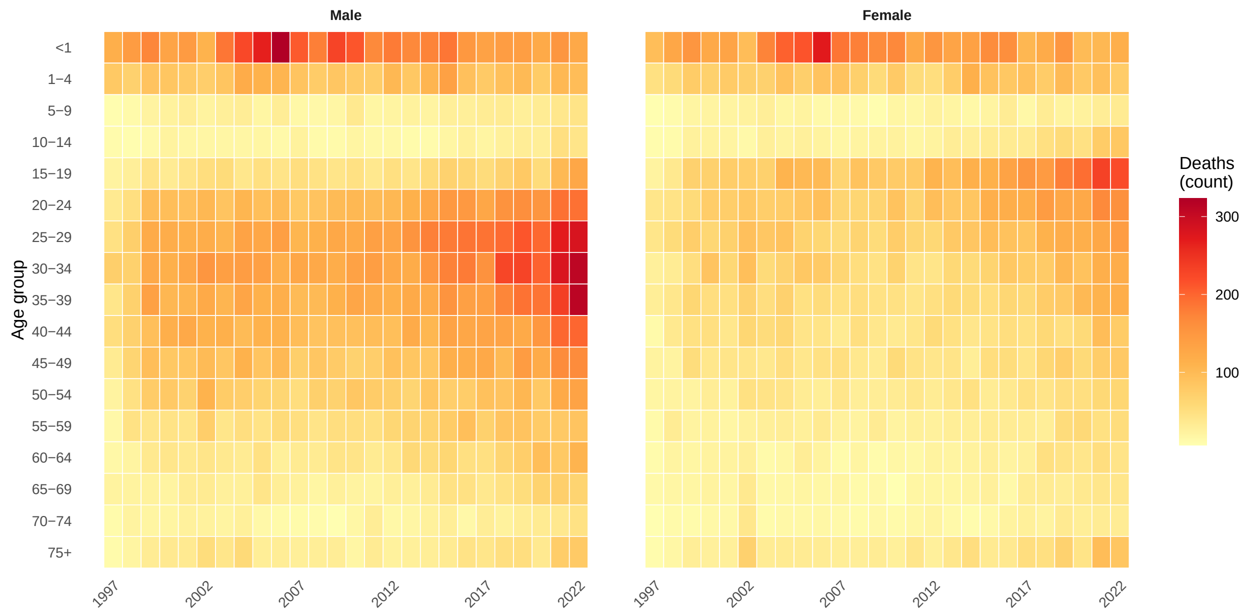

Figure 11: Fig. 11a. Annual poisoning deaths by age group, stratified by sex (raw count). Each panel's colour scale is shared, so darker cells can be compared directly between male and female panels.

Age × year heatmap — within-sex proportional share

Each cell is the **percentage of that sex's total deaths** in that year falling in that age band. Proportions are computed separately within each sex, so the panels show whether the age structure of male and female poisoning deaths has shifted over time — independently of the overall sex ratio in deaths.

EAPC by age group — male vs female comparison

Log-linear EAPC estimated separately for each age group × sex combination. Blue circles = males; red triangles = females; faded = p 0.05. Reading across a row for a given age band shows which sex is driving any trend — and whether both sexes are changing in the same direction and magnitude.

Geographic distribution

Counts by province of **usual residence** (`ResProvince`), excluding records coded as "Outside SA" or "Unspecified".

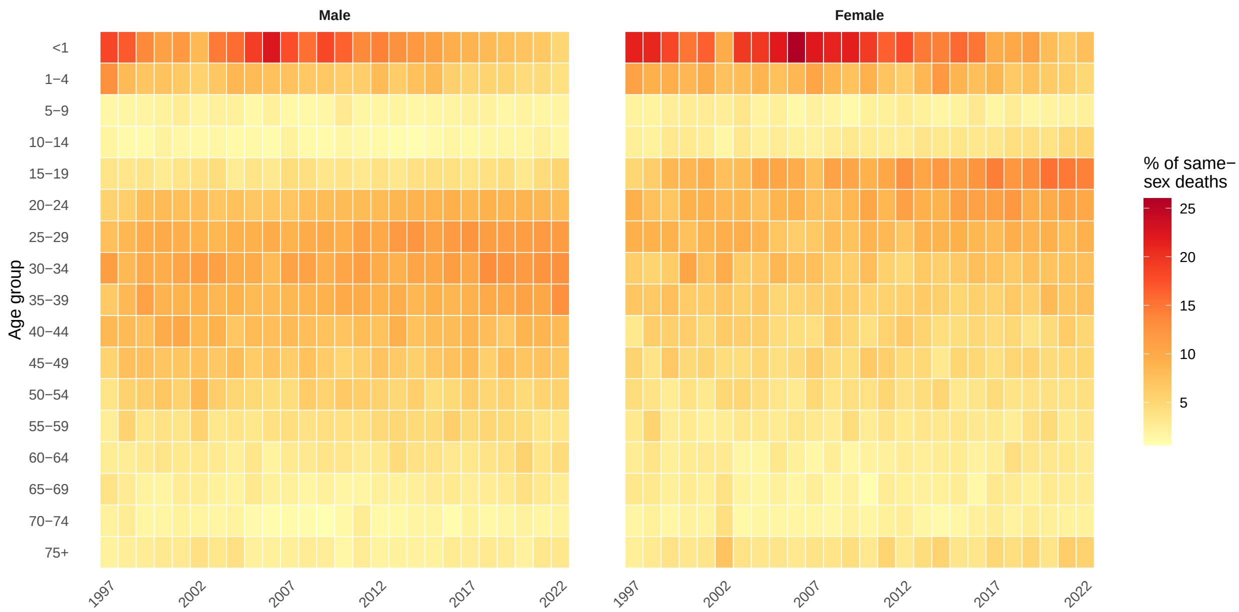

Figure 12: Fig. 11b. Proportion of annual poisoning deaths by age group within each sex. Each panel's columns sum to 100%.

District hot spots: last 3 years

Pooling the last 3 years stabilizes district-level crude rates and makes potential geographic hot spots easier to interpret than a single-year map.

Cause breakdown — ICD subcodes

The top 15 three-character ICD-10 subcodes within the poisoning block, ranked by total deaths across the study period.

Years of Life Lost (YLL)

YLL is computed as the sum of remaining life expectancy at age of death across all registered poisoning deaths with a known, valid age and documented sex (male or female). Individual ages are mapped to WHO standard remaining life expectancy using linear interpolation of the WHO reference life table (Murray 1994; LE at birth = 82.5 years). Records with unknown age or unknown/other sex are excluded from this section but remain in the death-count analyses above.

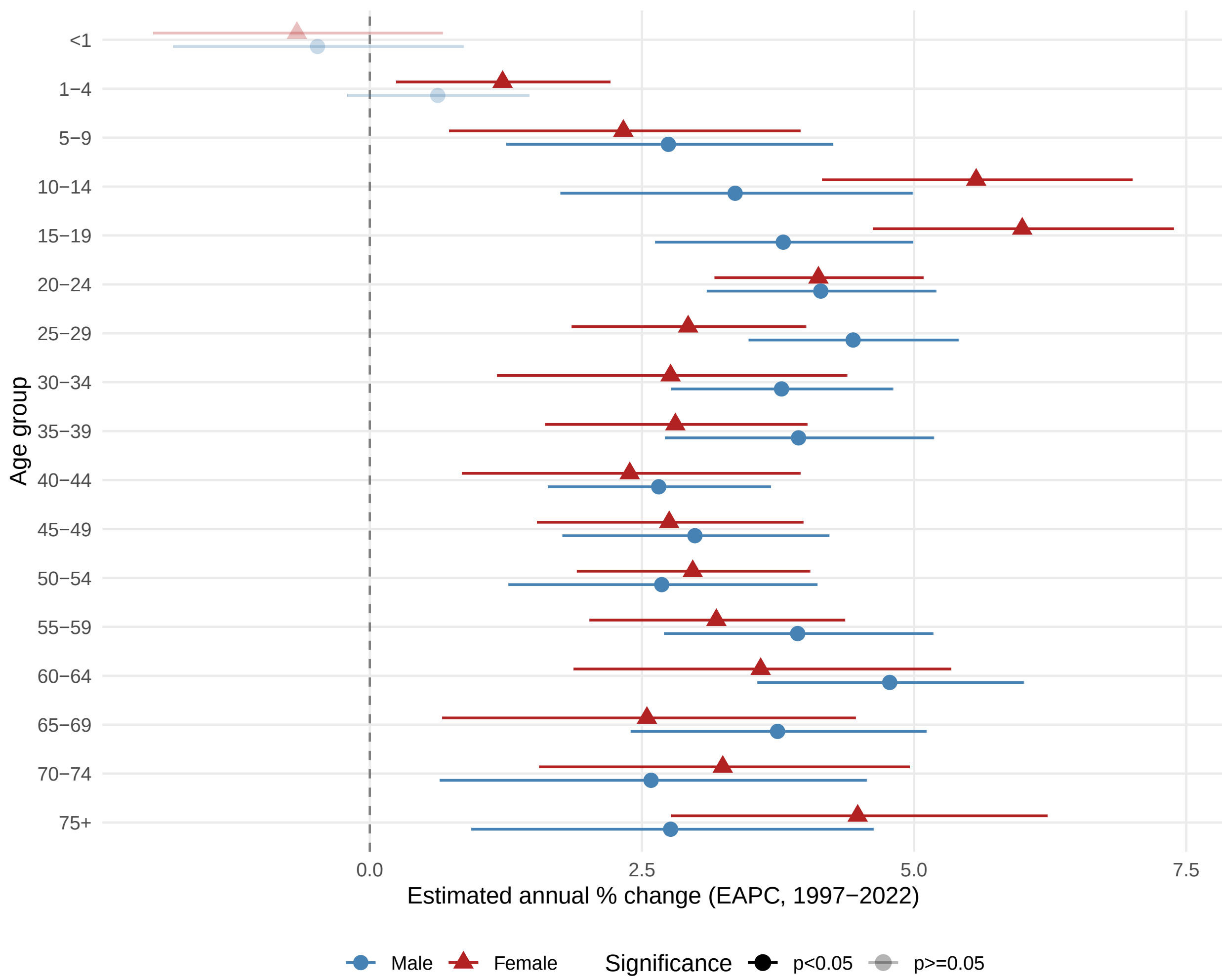

Figure 13: Fig. 12. EAPC by age group and sex with 95% CI. Blue circle = male; red triangle = female. Faded = non-significant (p 0.05). Points are horizontally dodged for legibility.

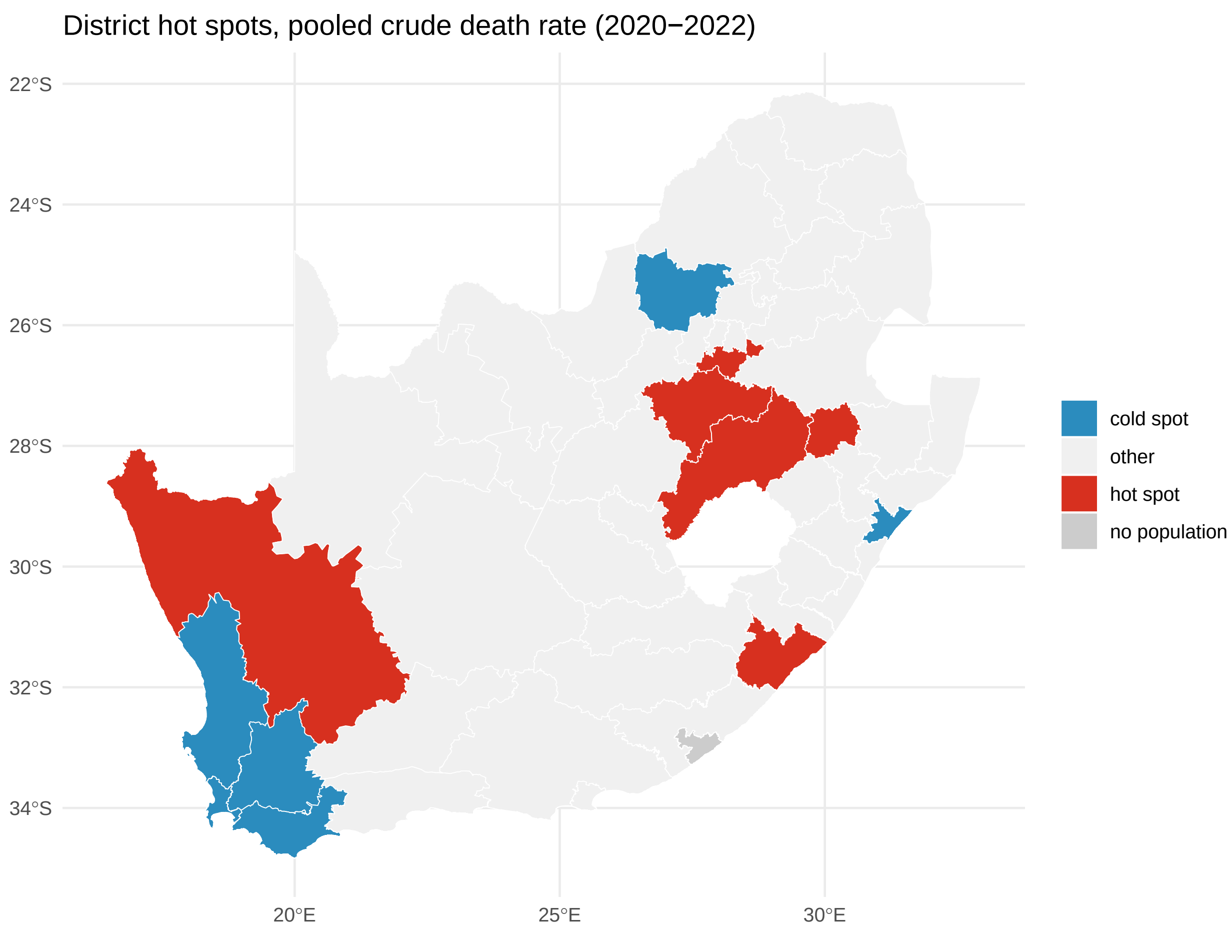

Figure 14: Fig. 7a. District hot spot map from pooled crude poisoning death rates over the last 3 years. Red = top decile of district crude rates; blue = bottom decile.

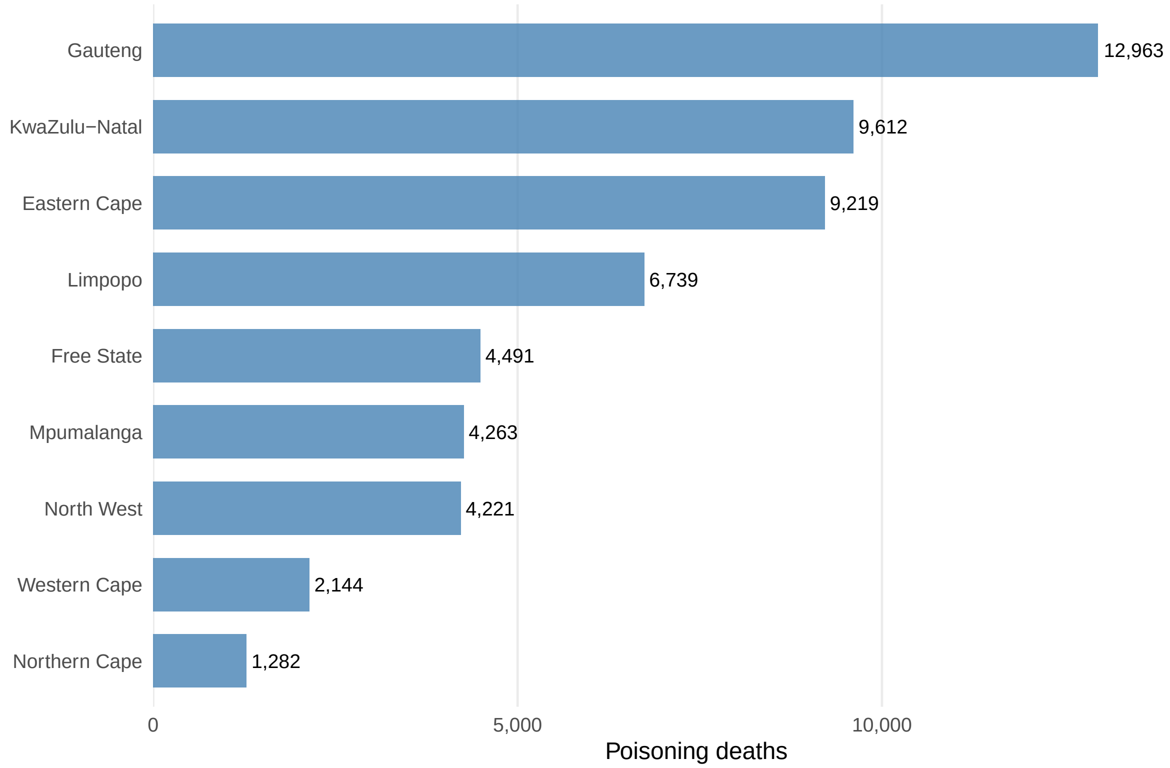

Figure 15: Fig. 7. Poisoning deaths by province of usual residence. Provinces ordered by count.

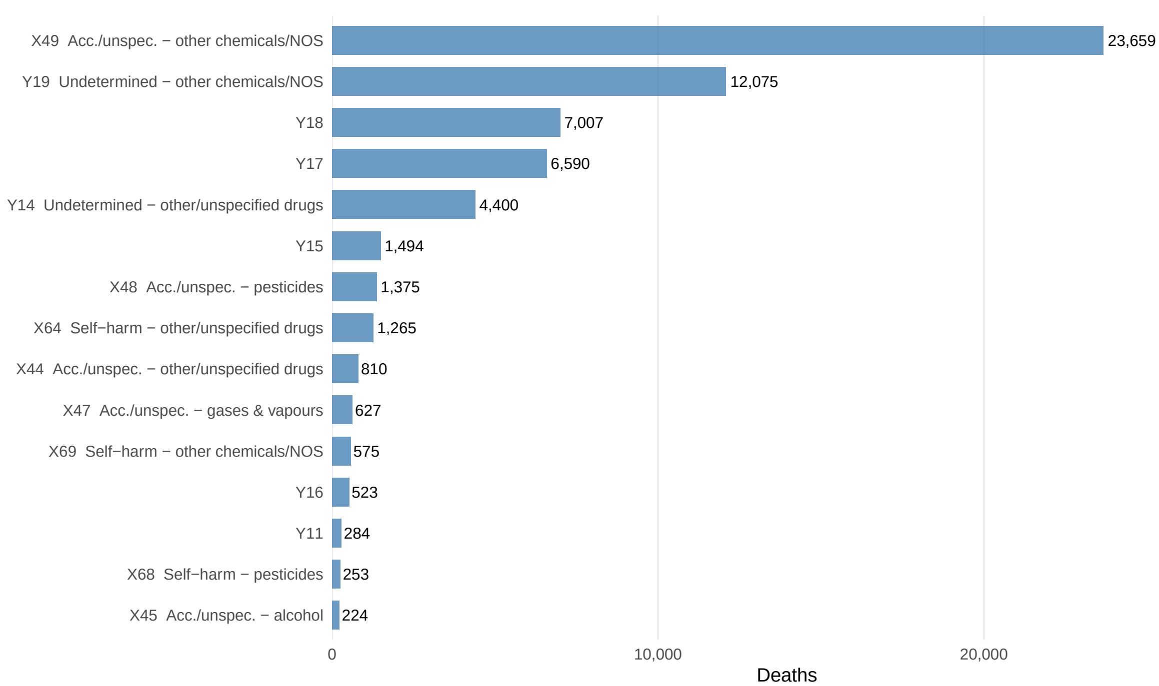

Figure 16: Fig. 8. Top 15 ICD-10 subcodes for poisoning deaths.

Reference life table

The WHO/GBD standard life table (Murray 1994) anchors YLL at a life expectancy of **82.5 years at birth**, providing a universal comparison benchmark independent of local South African life expectancy — which has fluctuated substantially over the study period due to HIV/AIDS and ART roll-out. Using a fixed standard means YLL reflects potential life lost relative to a best-practice survival frontier rather than current national survival. Standard values used for key ages: age 0 → 82.5 yr; age 20 → 62.8 yr; age 40 → 43.1 yr; age 60 → 24.3 yr; age 75 → 11.8 yr.

Summary

Table 4: Table 4. YLL by age group and sex — all poisoning deaths with valid age and kno (1997–2022). YLL computed using WHO standard life table (Murray 1994). 'YLL = mean standard life-years lost per registered death.

Age group	Male deaths	Male YLL	Male YLL/death	Female deaths	Female YLL	Female YLL/death
<1	4,502	371,460	82.5	3,874	319,644	82.5
1-4	2,394	193,555	80.8	2,023	163,550	80.8
5-9	693	52,689	76.0	565	42,904	75.9
10-14	558	39,411	70.6	832	58,263	70.0
15-19	1,424	93,150	65.4	2,825	185,639	65.7
20-24	2,969	179,908	60.6	2,406	146,553	60.9
25-29	3,904	217,935	55.8	2,143	119,814	55.9
30-34	4,018	204,984	51.0	1,820	92,838	51.0
35-39	3,518	162,460	46.2	1,598	73,801	46.2
40-44	3,009	124,195	41.3	1,304	53,780	41.2
45-49	2,558	93,184	36.4	1,252	45,579	36.4
50-54	2,046	64,861	31.7	1,014	32,213	31.8
55-59	1,581	42,754	27.0	835	22,682	27.2
60-64	1,285	29,078	22.6	684	15,447	22.6
65-69	944	17,356	18.4	610	11,118	18.2
70-74	611	8,704	14.2	521	7,376	14.2
75+	1,005	7,099	7.1	1,074	7,371	6.9

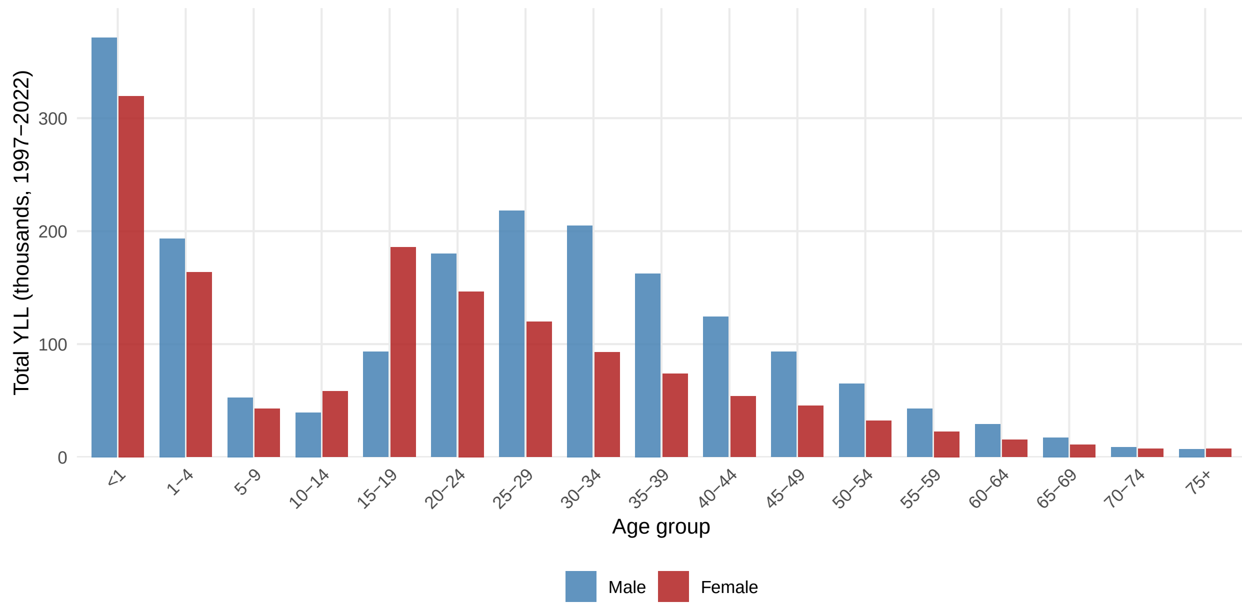

Figure 17: Fig. 13. Total YLL by age group, stratified by sex (1997–2022). Blue = male; red = female. YLL computed from individual registered ages using the WHO standard life table.

Total YLL by age group and sex

Mean YLL per death by age group

YLL per death declines monotonically with age because older decedents have fewer remaining standard life-years. The pattern reveals where the **per-death burden** is highest — typically in younger age groups — even when older groups contribute more deaths in absolute terms.

Annual YLL trend by sex

YLL heatmap — age group × year

Death institution / place of death

The `DeathInst` field on the South African death notification form records where the death occurred. Values are: **Hospital** (1), **Emergency room / Outpatient** (2), **Dead on arrival** (3), **Nursing home** (4), **Home** (5), **Other** (6), **Unknown** (8), and **Unspecified** (9). For the broad grouping below, categories 1, 2, and 4 are combined as **Health facility**; category 3 (**Dead on arrival**) is kept separate because it implies the death occurred outside a facility but was certified at one; categories 5 and 6 form **Out of facility**.

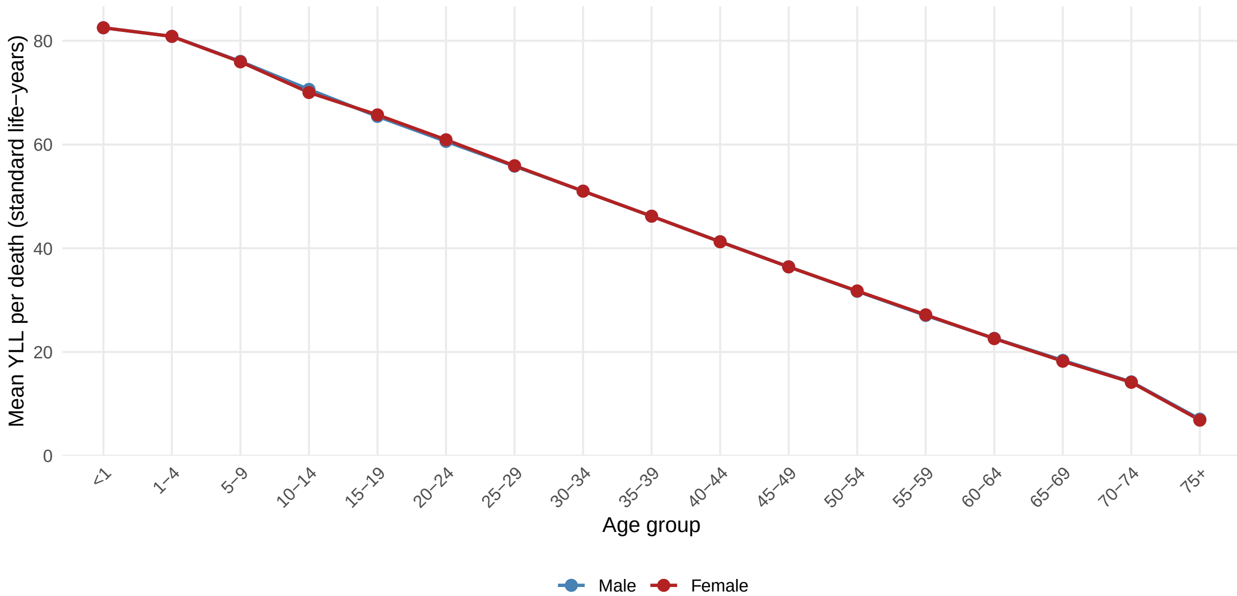

Figure 18: Fig. 14. Mean YLL per registered poisoning death by age group and sex. Reflects the WHO standard remaining life expectancy applied at each recorded age.

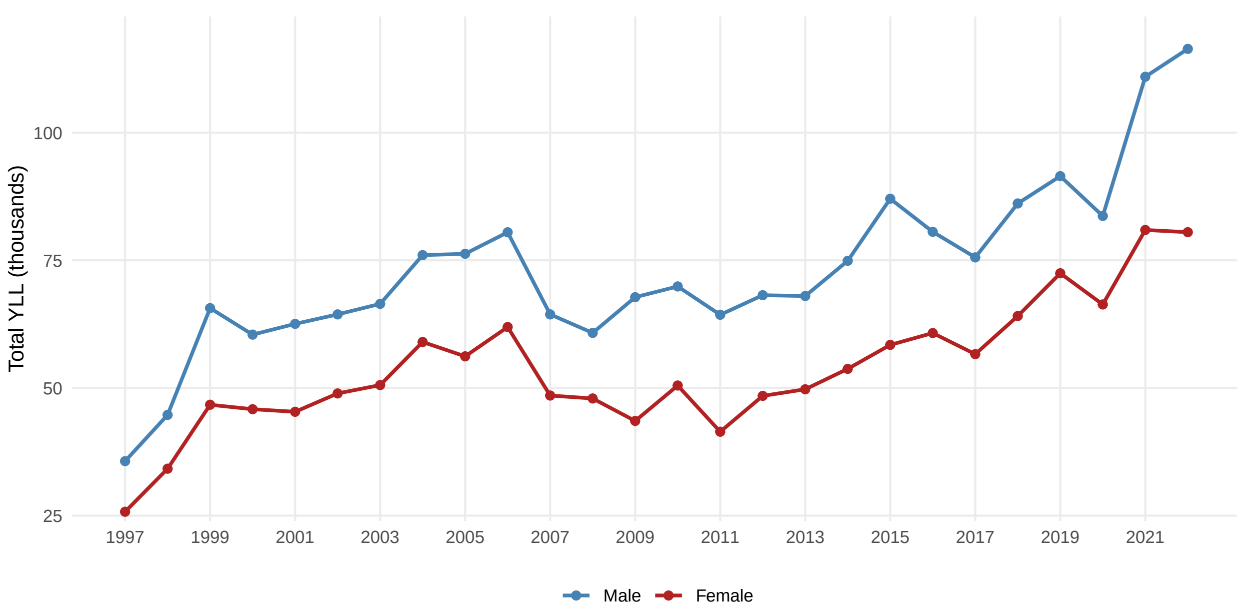

Figure 19: Fig. 15. Annual total YLL from poisoning deaths by sex (1997–2022). Blue = male; red = female.

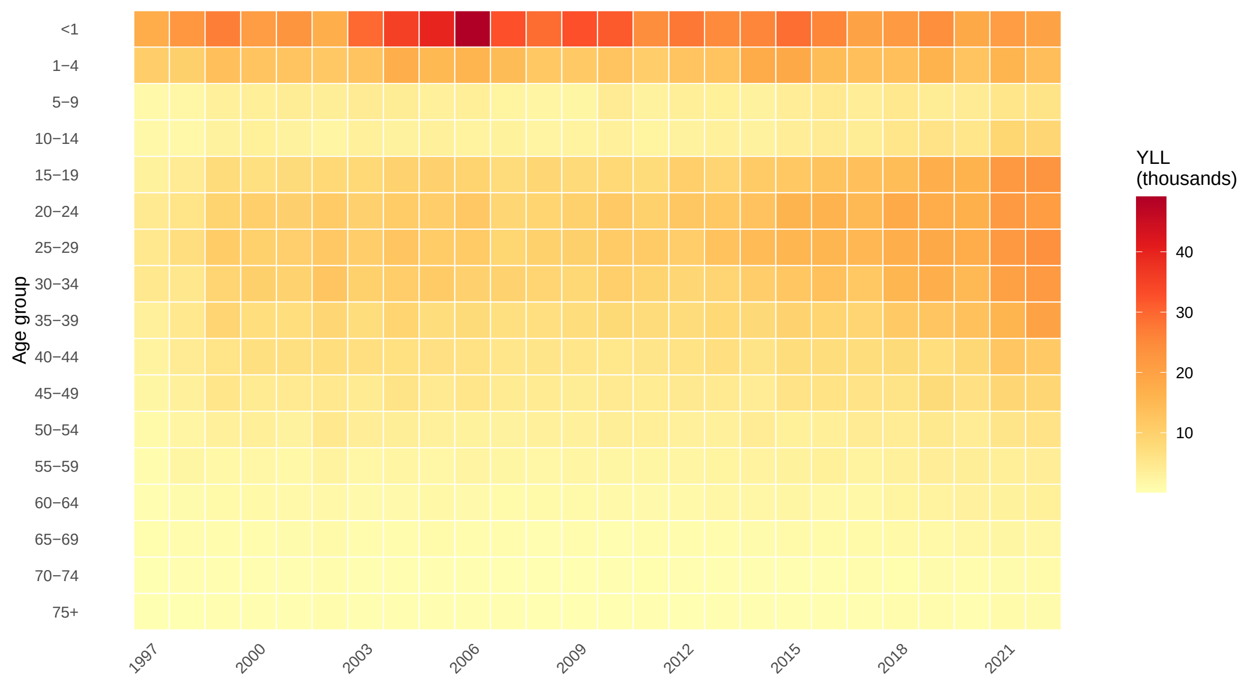

Figure 20: Fig. 16. YLL by age group and calendar year (heatmap). Darker cells = higher YLL. Combines the effect of death counts and age-at-death distribution.

Unspecified records (code 9)

A substantial proportion of records carry code 9 (Unspecified). This likely reflects incomplete form completion rather than a genuinely unknown setting, and mirrors the broader data-quality limitations of the vital registration system. These records are retained in the tables below but excluded from the bar charts to avoid distorting the proportions of known categories. Their exclusion is reported in the figure footnotes.

X40–X49: distribution by death institution

The chart below shows where accidental/unspecified poisoning deaths (X40–X49) occurred, excluding the Unknown and Unspecified categories. The "Unspecified" share is shown in the table.

Table 5: Table 5. X40–X49 poisoning deaths by death institution (known records only, excluding Unknown/Unspecified).

Death institution	Facility group	N	% of X40-49
Hospital	Health facility	18,611	68.8%
Home	Out of facility	1,435	5.3%
Emergency room/Outpatient	Health facility	1,035	3.8%
Dead on arrival	Dead on arrival	497	1.8%

Death institution	Facility group	N	% of X40-49
Other	Out of facility	481	1.8%
Nursing home	Health facility	59	0.2%

Intent composition within each facility setting

The proportional stacked bars show how the intent mix varies across facility settings. Because "Accidental/unspec." is the ICD-10 default when manner is not stated, settings where non-forensic clinicians certify deaths (e.g. hospital wards) are expected to show the highest proportional contribution from this category.

Table 6: Table 6. Poisoning deaths by facility setting and intent (all records, including Unknown/Unspecified). '% Acc./unspec.' = proportion of that setting's deaths coded X40–X49.

Facility setting	Accidental/unspec.	Self-harm	Assault	Undetermined	Total
Health facility	19,705	1,137	4	11,099	31,945
Dead on arrival	497	148	0	1,480	2,125
Out of facility	1,916	487	0	7,869	10,272
Unknown/unspecified	4,916	830	2	12,309	18,057

Next steps

- Add population denominators to compute age-standardised mortality rates.
- Stratify provincial trends by intent.
- Link to external toxicology data (e.g. NPIS calls, emergency department presentations) for triangulation with IMS and NCoDV forensic data.
- Investigate the spike/dip years visible in the epicurve.
- Apply probabilistic redistribution of the "Accidental" category using forensic correction factors (e.g. from (3)) to obtain intent-adjusted estimates.

References

1. Groenewald P, Matzopoulos R, Afonso E, Bradshaw D. The importance of including manner of (injury) death on the death notification form. S Afr Med J. 2023;113(9):11–2. doi:10.7196/SAMJ.2023.v113i9.915 PubMed PMID: 37882128.

2. World Health Organization. International statistical classification of diseases and related health problems, 10th revision, volume 2: Instruction manual [Internet]. 5th ed. Geneva: World Health Organization; 2016. Available from: https://apps.who.int/classifications/icd10/browse/2016/en

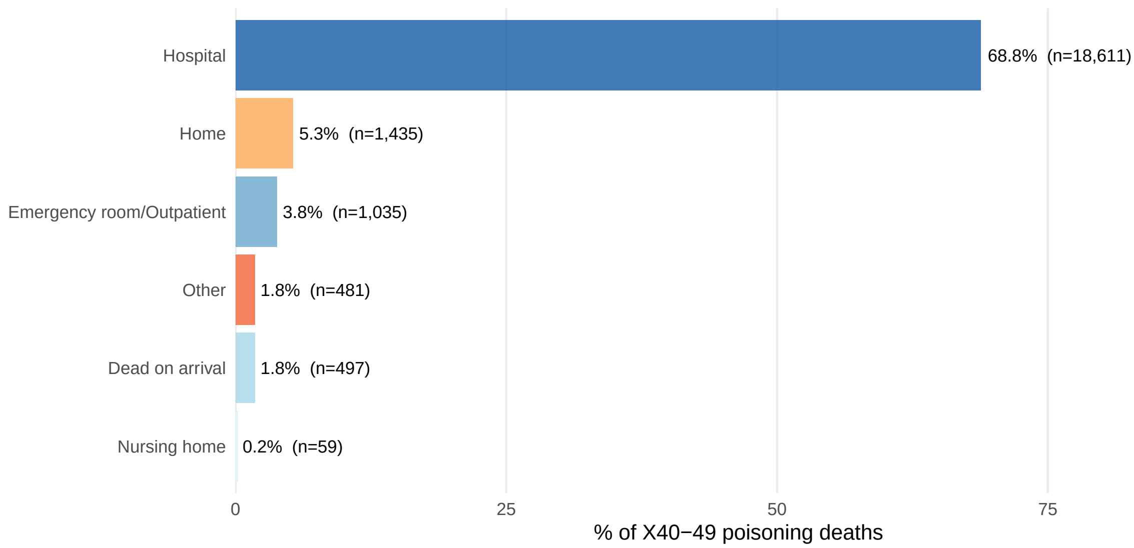

Figure 21: Fig. 17. Death institution for X40–X49 (accidental/unspecified) poisoning deaths. Records coded Unknown (8) or Unspecified (9) are excluded from this chart but included in the table below.

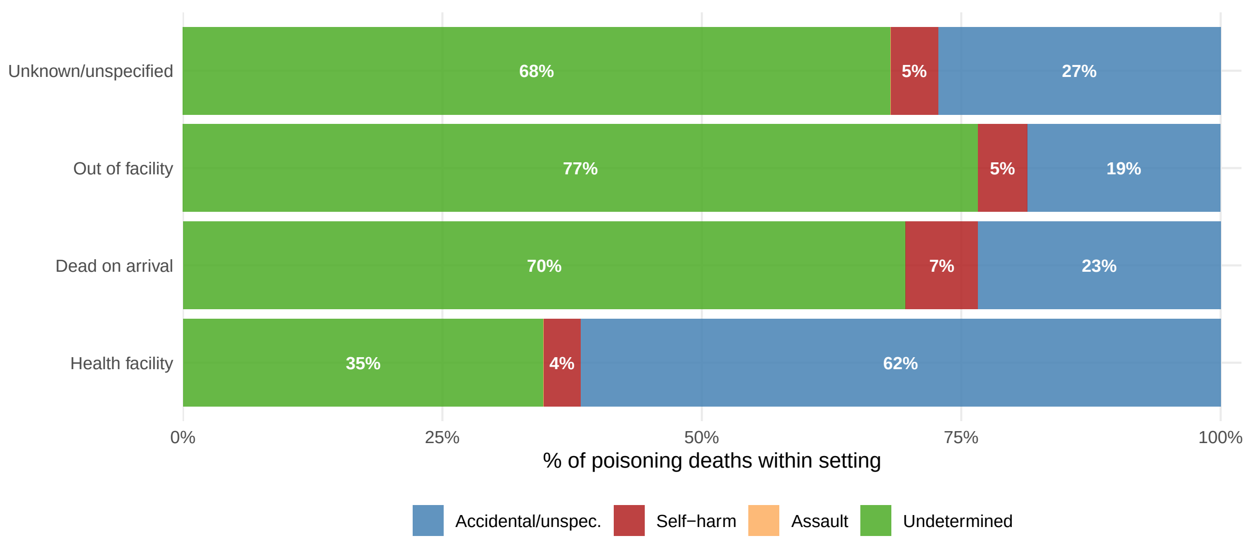

Figure 22: Fig. 18. Intent composition of poisoning deaths within each facility setting (100% stacked bars). 'Other' intent excluded. Percentages are within-setting shares.

3. Groenewald P, Kallis N, Holmgren C, Glass T, Anthony A, Maud P, et al. Further evidence of misclassification of the injury deaths in South Africa: When will the barriers to accurate injury death statistics be removed? S Afr Med J. 2023;113(9):30–5. doi:10.7196/SAMJ.2023.v113i9.836 PubMed PMID: 37882130; PubMed Central PMCID: PMC11017197.

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